Qijia Fan

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EDUCATION AND RESEARCH EXPERIENCE

Tsinghua University *Sept. 2025 – Present*
Ph.D. student of Computer Science, advised by Prof. Yifan Song.

Brown University *Aug. 2024 – Dec. 2024*
Research Intern.

Shanghai Jiao Tong University *Aug. 2021 – Jun. 2025*
Bachelor in Computer Science.
- A member of **ACM Honors Class**, an elite CS program in SJTU.
- GPA: 94.27/100, 4.11/4.3 | Ranking: 2/34

MY INTERESTS AND PUBLICATIONS

I am now interested in *theoretical computer science*, especially in *cryptography*. I am fond of using mathematical methods to design or analyze an algorithm or a protocol.

Two-Factor Authenticated Key Exchange with Enhanced Security from Post-Quantum Assumptions
***Qijia Fan**, Chenhao Bao, Xuanyu Shi, Shuai Han, and Shengli Liu. ESORICS 2025.*
We proposed a two-factor authenticated key exchange protocol that provides perfect forward security and mutual explicit authentication while protecting the privacy of biometrics and passwords.

Edabits Generation with Quasi-Linear Communication Complexity
*Alexander Bienstock, Daniel Escudero, **Qijia Fan**, Fengrun Liu, and Yifan Song. IACR ePrint 2026/2482.*
We develop an edaBits generation protocol with quasi-linear communication complexity, facilitating efficient conversions between arithmetic and binary computation modes in secure multi-party computation.

AWARDS AND SCHOLARSHIPS

- **National Scholarship**, Ministry of Education of the People's Republic of China *2023, 2024*
- **Outstanding Scholarship for Undergraduates, Grade A** (the highest level, top 5% of the school) *2023, 2024*
- **Irving T. Ho Memorial Scholarship** *2022*
- **Zhiyuan Hanying Ju Hua Scholarship** *2023*

TEACHING EXPERIENCE

Modern Algebra *Spring, 2024*
Teaching Assistant with Prof. Shengli Liu
I helped students to get a better understanding of group, ring and field, and correct their homework. I also helped prepare exam questions.

Principle and Practice of Computer Algorithms *Summer, 2023*
Teaching Assistant
I helped students to understand the Raft Consensus Algorithm and implement the algorithm.

Data Structures *Spring, 2023*
Teaching Assistant with Prof. Huiyu Weng
I helped students to implement mergeable heaps and analyze the time complexity of some self-adjusting heaps (like skew heaps, pairing heaps, Fibonacci heaps, etc.) with amortized analysis method.

SKILLS

Languages: C/C++, Go, Java, Verilog, Coq, LaTeX, Python.